Robotics project proposal: measured slip and fine terrain for wheeled robots on Gulf dunes

Status: PROPOSAL. No agreement, partnership, funding or endorsement exists with any institution, and none is implied. Author: Huda Mueen, `hudasol/dune-rover`. Generated from `docs/src/proposal_robotics_template.md` by `scripts/build_report.py`; every number comes from a result file or `research/evidence.csv` and is traced in `docs/NUMBER_LEDGER.csv`. Nothing has been sent to anyone. No price or cost is given: the resources below are what a partner would be asked to provide. Heat and dust limits for people and electronics are covered in the separate repository `hudasol/dhill`.

Labels: **PUBLISHED** (read in a retrieved source), **SIMULATED BY ME** (output of this repository's code), **PLANNED** (not run).

1. Problem

A small wheeled robot on a dune slope can lose most of its forward progress to slip, and the slope at which it stops depends on the vehicle, its wheels and the sand. Route planning on Gulf dune fields needs (i) a slip law that holds for the vehicle on that sand and (ii) a terrain model fine enough to see slip faces. Neither exists in open form for Arabian dunes: no published slip against slope dataset was found for any wheeled robot on Arabian dune sand, and open elevation data are about 30 m.

2. What the desk study established

- **Evidence base (PUBLISHED):** 510 verified rows from 176 sources, each re checked against the full text, with a measured extraction error rate (7.5 % of the current rows had a material error before correction; every one corrected).
- **Slip models do not transfer (SIMULATED BY ME against PUBLISHED curves):** on 49 held out published curves, the best of three model families (exponential, Mohr Coulomb, logistic) predicted a study it was not calibrated on with a median slip error of 0.20 (bar fixed in advance: 0.15); a fourth, Bekker Wong, failed on the 8 first round curves it applies to (0.20) and did worse on the 33 curves of the expanded set (0.41). Two families built on the granular scaling laws did worse (0.31 to 0.34).

- **A simulator calibrated on a field curve gave no usable steep slope prediction (SIMULATED BY ME):** calibrated on the Dumont Dunes curve up to 12.5 degrees, Project Chrono's Soil Contact Model was speed sensitive at 15 degrees and not numerically stable at 17.5 and 20 degrees (post hoc grid and speed checks), so the steep slopes still need measurement.
- **30 m terrain data are not enough (SIMULATED BY ME on PUBLISHED lidar):** on 6 dune fields, a 30 m map labelled TRAVERSABLE 36 % to 80 % of the cells that contain a steep face; at 10 m, 7 % to 33 %.
- **Arabian dunes from space (PUBLISHED ICESat-2 data, SIMULATED BY ME scoring):** Over Liwa (A), the Rub' al Khali megadunes (B) and the linear dunes (C), 71 %, 47 % and 67 % of the ICESat-2 segments steeper than 28 degrees fell in cells the 30 m map calls TRAVERSABLE, against 93 %, 92 % and 98 % of all segments; keeping only steep crossings confirmed by a neighbouring segment and no steeper than 35 degrees (post hoc), 52 %, 29 % and 49 %. The ICESat-2 method failed its own validation against lidar, so these are indications, not a validation.
- **A closed loop in simulation (SIMULATED BY ME, the true rover is itself a model):** online slip estimation from odometry and position fixes had a median error of about 0.029, which follows from the assumed position noise. Over 5 independent slip fields and 40 pairs per site (post hoc, after two reviews), replanning on that estimate changed the entrapped share on 30 m naive routes by minus 6.0 points at White Sands (95 % interval minus 20.5 to plus 8.0) and plus 2.0 at Kelso (minus 12.5 to plus 16.0) against a control that stops and retries without replanning: no effect distinguishable from the retry control. The loop runs as ROS 2 nodes (one smoke test, on a pair that also arrives without replanning).

3. The product the project builds on: the Dune Rover Planner (SIMULATED BY ME)

The desk study is packaged as a working planning tool (`docs/PLANNER.md`, report section 15). From a dune elevation model and a vehicle profile it produces a slip face risk map, a three class map (TRAVERSABLE, BLOCKED, UNKNOWN), routes under three policies with their risks, a GeoTIFF class map, GeoJSON routes and an HTML report. Its resolution gate carries the main finding: ground coarser than 10 m is never reported as verified traversable, because a 30 m map called 36 % to 80 % of the face cells TRAVERSABLE on 6 lidar dune fields. Its vehicle profiles carry the second: the tool assumes nothing about the rover until it is calibrated from the rover's own slope runs (`dunerover vehicle calibrate`). An interactive demonstration on 4 lidar dune fields is in `demo/web/dune_rover_planner.html`. The tool exports a Nav2 occupancy map, QGroundControl missions and KML, and runs as a ROS 2 node. A preregistered benchmark against baselines (report section 16, `paper/paper.pdf`) shows why the fine survey matters: with every hazard

layer computable from a 30 m map, mission success stayed at most 11 %, while a hard gate on a face map made from fine data still found face free routes for 83 % of pairs on the same 30 m grid (post hoc).

What the project adds is what the tool cannot produce by itself: a fine survey of an Arabian dune (WP5), so the planner runs in its resolved mode on Arabian terrain, and a measured slip curve for the rover (WP4), so its routes rest on the vehicle rather than on assumptions. The project ends with the planner calibrated to the partner's rover and checked on the partner's dune.

4. Technology readiness, stated honestly

The levels below use the common nine step technology readiness scale only as a shorthand; no assessment by any body has been made.

Component	Readiness	Evidence
Slip model for a vehicle on Arabian sand	Concept only (level 2): no model transferred between published studies, and no Arabian measurement exists	Stages 15, 21, 22
Slip face detector and three class map	Proof of concept against lidar of dune fields outside Arabia (level 3): works at 10 m or finer, fails at 30 m	Stages 12, 20
Map on Arabian dunes	Compared only with satellite laser profiles, with a method that did not pass its own validation bar and a noisy steep segment set	Stage 19
Online slip estimation and replanning	Software in simulation against a modelled rover, including ROS 2 nodes (level 2 to 3); no benefit of replanning shown against a retry control	Stage 23
Field protocol, data sheets, analysis and ingest	Written and tested on synthetic data (level 3); pass or fail rule still an open decision	`docs/FIELD_VALIDATION_PLAN.md`, `scripts/ingest_field_day.py`
Dune Rover Planner (integrated tool)	Installable command line tool and Python API, tested on synthetic inputs and demonstrated on lidar outside Arabia (level 3, resting on the detector and the map); its routing did not pass its registered test at 30 m (stage 17), no hazard layer made 30 m routes meaningfully safer in the benchmark of stage 26, and no vehicle is calibrated yet	`docs/PLANNER.md`, `tests/test_planner.py`

5. Objectives

1. Measure the slip against slope curve of one small wheeled robot on one Gulf dune with the sand characterised (grain size, density, moisture, angle of repose, friction angle).
2. Survey the test dune at 10 cm (drone photogrammetry), and score the slip face detector and the three class map on Arabian terrain at 1, 5, 10 and 30 m.
3. Test whether a model calibrated on gentle slopes predicts the steep slopes, against predictions committed before the steep runs.
4. Measure online slip estimation on the real rover on the surveyed dune, with a safety driver, and test whether replanning on it lowers entrapment against a control that stops and retries.
5. Deliver a documented dataset and a manuscript revised with the field results.

6. Work packages

Durations are planning estimates (PLANNED), in weeks from the start of the project.

Work package	Content	Weeks	Depends on	Deli
WP1 Preparation	Site access in writing, permissions for any drone flight, safety plan, rover choice (`docs/ROVER_OPTIONS.md`), decision on the field rule (question 7), preregistration submitted (`docs/OSF_PREREGISTRATION_DRAFT.md`)	1 to 4	none	Sign prer
WP2 Rover and instruments	Wheel encoders, RTK or fixed position logging, IMU, effective radius calibration, closed loop software from `demo/ros2/` on the rover computer	3 to 6	WP1	Ben
WP3 Sand tests	Grain size, density, moisture, angle of repose at every station; direct shear if a laboratory allows (`docs/SAND_LAB_PROTOCOL.md`)	5 to 8	WP1	Soil
WP4 Pilot and slope ladder	12 pilot runs, pilot scatter, then the main ladder with predictions committed before the steep bins (`field_kit/`)	6 to 9	WP2, WP3	Run (`sc`
WP5 Drone survey	Survey of the test dune at about 10 cm with ground control and check points (`docs/DRONE_SURVEY_PROTOCOL.md`); detector and map scored as in stage 20	6 to 9	WP1	Elev scor
WP6 Closed loop trial	Route following on the surveyed dune with online slip estimation; replanning tested against a stop and retry control (in simulation it showed no benefit over that control); safety driver and recovery	10 to 12	WP4, WP5	Esti outc
WP7 Analysis, paper and calibrated planner	Every hit and miss recorded; manuscript revised; external review; the planner given the rover's measured profile and run on the surveyed dune	12 to 16	WP4 to WP6	Rev plar

7. Resources the partner is asked to provide (no prices; nothing is requested from the author's own funds)

Need
A named supervisor and a safety review
Access to one dune site with permission in writing
One small wheeled ground vehicle (5 to 25 kg, four wheels, wheels at least 150 mm in diameter, known mass) or support
RTK GNSS (rover and base) or a total station
A drone flight by an authorised operator, with ground control targets
Laboratory access for sieving, density and, if possible, direct shear
A recovery vehicle and an experienced sand driver
One outside reviewer in terramechanics or field robotics

8. Success and failure criteria

Criterion	Tied to	Counts as success	Counts as failure
Field data complete	WP4	Every planned run logged, including entrapments and aborted runs, with soil tests at every station	Missing runs without a written reason
Predictions scored	`docs/predictions/` and the committed rule	Every dated prediction scored as hit or miss, whatever the result	Any prediction changed after a run
Arabian map test	Stages 12, 19, 20	Detector AUC and the 30 m miss rate reported for the surveyed dune	No survey, or a survey without check points
Closed loop on the rover	Stage 23	Entrapment share with and without replanning reported on the same routes	Results reported only for routes that succeeded

Success means complete data and an honest comparison, not a model that passes.

9. Risk and safety register

Risk	Likelihood	Impact	Mitigation
Rover entrapment or rollover on a steep face	High	Medium	Tether and recovery line, ab (`docs/FIELD_DAY_CHEC` no person downslope, steep
Heat illness of the team	Medium	High	Work in the early morning, v shade, limits from `hudaso`

Risk	Likelihood	Impact	Mitigation
Drone flight without the right permission	Medium	High	Fly only through an operato the authorisation; no flight c
Run to run scatter too large to discriminate	Medium	Medium	Pilot first; the analysis flags uninformative experiment
The pass or fail rule cannot separate a correct from a collapse model	High	Medium	Decide question 7 before th report predicted against obs main result
Sand properties change during the day (crust, moisture)	Medium	Medium	Soil tests at every station ar recorded
Single vehicle, single site	Certain	Medium	Results stated for that vehic sand and speed only
Implied endorsement	Low	High	Every document states that agreement exists unless on

10. What this proposal does not claim

It does not claim a validated slip model, a traversable route, a cost or schedule beyond the planning estimates above, or any institutional support.